WEATHERLAB

BS 2 Linz · Class 4bITS

IoT Environmental Monitoring System

Smart weather station & indoor air quality monitoring: sensor data, local web server with touch display and a 3-day forecast.

PROJECT GOAL

We are building a self-contained IoT weather station that measures **room temperature, humidity, pressure and air quality** live and also shows a **3-day weather forecast**. The Raspberry Pi hosts the web GUI directly on the 7-inch touch screen. A dedicated VM provides admin guides and the project documentation.

SYSTEM FLOW

BME680
Temperature, air quality, humidity, pressure

→

Raspberry Pi
Data processing, local web server, logging

←

OpenWeather API
External 3-day forecast

→

7" Touch-GUI
Interactive live view

Sensor values are read at fixed intervals and stored locally (logging). The documentation VM runs independently as its own web server.

MUST Requirements — MANDATORY

- **Sensor:** BME680 wired to Raspberry Pi, continuous readings
- **Web GUI:** live display on the 7" touch screen
- **API:** 3-day forecast via OpenWeather
- **VM portal:** docs, hardware intro, admin guides

CAN Requirements — OPTIONAL

- **Telegram bot:** storm and temperature-drop alerts
- **LED indicator:** colour signal for air quality
- **Smartphone webview** of the forecast
- **Smart home hub:** control of further devices

PROJECT TEAM

Andrea Roithmeier	Project Lead
David Sageder	Vice Lead, Hardware
Fabian Seelig	API & Backend
Julian Weißböck	Documentation
Matthias Winklehner	Web, time tracking

Supervision: David Steinegger

PROJECT MANAGEMENT

Organigram with role descriptions

CPM network plan in ProjectLibre (critical path)

Weekly protocols, signed

Time tracking with Kimai

Timeline: Sep 15 – Nov 10, 2026
25 tasks in 5 groups: Raspberry Pi, documentation, web server, organisational matters

MATERIAL COSTS

€ 137.16

BME680 26.29 · display 65.87 · microSD 30.00 · accessories 15.00 (Raspberry Pi provided)

PERSONNEL COSTS

€ 3,000.00

5 people × 30 h × € 20

TOTAL COST

€ 3,137.16